

# Software Requirements Specification (SRS)

## Code Review Analytics System Based on Machine Learning

### Document Control

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## Glossary

| Term   | Definition                                   |
|--------|----------------------------------------------|
| AI     | Artificial Intelligence                      |
| API    | Application Programming Interface            |
| GDPR   | General Data Protection Regulation           |
| LMIS   | Labor Market Information System              |
| MoL    | Ministry of Labor                            |
| Enabel | Belgian agency for international cooperation |
| NLP    | Natural Language Processing                  |
| PEF    | Palestinian Employment Fund                  |
| RBAC   | Role-Based Access Control                    |
| RTO    | Recovery Time Objective                      |
| RPO    | Recovery Point Objective                     |
| SLA    | Service Level Agreement                      |
| SRS    | System Requirements Specification            |
| ToR    | Terms of Reference                           |
| WCAG   | Web Content Accessibility Guidelines         |
| YEP    | Youth Economic Empowerment in Palestine      |

## 1. Introduction

### 1.1. Purpose

The purpose of this Software Requirements Specification (SRS) document is to provide a detailed description of the requirements for the **Code Review Analytics System Based on Machine Learning**. This document defines the functional and non-functional requirements of the system, including its features, constraints, and expected behavior. It serves as a reference for the development team, project guide, and evaluators to understand the scope and objectives of the proposed system.

The primary objective of the system is to analyze software code review data and evaluate the effectiveness of different code review practices such as peer review, pull-based review, and tool-assisted review. The system collects information related to defects, review time, and severity, and applies analytical and machine learning techniques to identify patterns and predict defect severity. Based on these inputs, the system computes a Weighted Code Quality Score (WQS) to provide a quantitative measure of software quality.

This document ensures that all stakeholders share a common understanding of the system requirements and helps guide the design, development, testing, and evaluation phases of the project. It also acts as a baseline for validating whether the developed system meets the intended goals of improving software quality through systematic and data-driven evaluation of code review practices.

### 1.2. Document Conventions

This document follows these conventions:

| Term          | Description                                                                                                          |
|---------------|----------------------------------------------------------------------------------------------------------------------|
| <b>SHALL</b>  | Refers to a mandatory requirement that must be implemented in the current phase of the Code Review Analytics System. |
| <b>SHOULD</b> | Indicates a recommended requirement that may be considered for future improvements or scalability.                   |
| <b>MAY</b>    | Refers to optional features that can be implemented in later phases if required.                                     |
| <b>TBD</b>    | To Be Determined; indicates information that will be finalized in future versions of the document.                   |
| <b>Note</b>   | Provides additional clarification or explanatory information related to system requirements.                         |

Requirements are categorized as follows:

| Requirement Number | Description                 |
|--------------------|-----------------------------|
| <b>FR-XX</b>       | Functional Requirements     |
| <b>NFR-XX</b>      | Non-Functional Requirements |
| <b>IR-XX</b>       | Interface Requirements      |
| <b>DR-XX</b>       | Data Requirements           |
| <b>SR-XX</b>       | Security Requirements       |

### 1.3. Intended Audience

This document is intended for the following stakeholders:

| Stakeholder                       | Role                                                                                      |
|-----------------------------------|-------------------------------------------------------------------------------------------|
| Project Guide / Faculty           | Will review, validate, and approve the project requirements and overall system design.    |
| Development Team                  | Responsible for designing, implementing, and testing the Code Review Analytics System.    |
| Software Engineering Evaluators   | Will assess the effectiveness, documentation, and quality of the implemented system.      |
| Students / Developers (End Users) | Will use the system to perform code review analysis and understand review effectiveness.  |
| System Administrator              | Will maintain the system environment and manage required software tools.                  |
| Research Review Panel             | Will analyze the empirical results and evaluate the research contribution of the project. |

#### 1.4. Project Scope

The Code Review Analytics System Based on Machine Learning is developed as part of the Software Engineering academic curriculum to analyze and evaluate different code review practices in a controlled environment. The system functions as an analytical framework that applies peer review, pull-based review, and tool-assisted static analysis on a sample software application. It collects review metrics, classifies defects, and computes a Weighted Code Quality Score (WQS) to enable objective comparison of review techniques. The system is limited to console-based implementation and is intended for academic evaluation and research purposes.

#### In Scope

The system will include the following key components:

**Sample Software System:** Development of a small software application that will be used for performing code review experiments.

**Code Review Management:** Support for conducting peer review, pull-based review, and tool-assisted review.

**Defect Identification and Classification:** Recording and categorizing defects based on type and severity (Critical, Major, Minor).

**Review Data Collection:** Storage of review metrics such as number of defects, review time, and reviewer comments.

**Weighted Code Quality Score (WQS) Calculation:** Implementation of a quantitative metric to evaluate software quality.

**Comparative Analysis Module:** Comparison of different code review techniques based on effectiveness and effort.

**Reporting and Visualization:** Generation of reports, charts, and graphs for analysis results.

**Basic User Interface:** Simple interface for entering review data and viewing results.

## Out of Scope

The following items are explicitly excluded from this phase of the project:

- Deployment in a real industrial environment.
- Integration with large-scale enterprise repositories or proprietary systems.
- Development of a mobile application.
- Full automation of code review using advanced AI models.
- Long-term system maintenance and production support.
- Integration with external third-party platforms not specified in this document.

## 1.5. References

This SRS is based on the following documents and sources:

- 1) IEEE Std 830-1998, *IEEE Recommended Practice for Software Requirements Specifications*.
- 2) Pressman, R. S., *Software Engineering: A Practitioner's Approach*, McGraw-Hill Education.
- 3) Sommerville, Ian, *Software Engineering*, 10th Edition, Pearson Education.
- 4) Git Documentation, Version Control and Pull Request Workflow.
- 5) Pylint Documentation, Static Code Analysis Tool for Python.
- 6) Scikit-learn Documentation, Machine Learning Library for Python.
- 7) Python Official Documentation, Python Software Foundation.

## 2. Overall Description

### 2.1. Product Perspective

The Code Review Analytics System Based on Machine Learning is a new system developed to analyze and evaluate the effectiveness of different software code review practices. The system will operate as a standalone analytics platform designed to support experimental evaluation of peer review, pull-based review, and tool-assisted review techniques in an academic environment.

The proposed system is intended to become a foundational component for future research and advanced software quality assessment frameworks. It will help in understanding how different review approaches impact software maintainability, defect detection, and overall code quality.

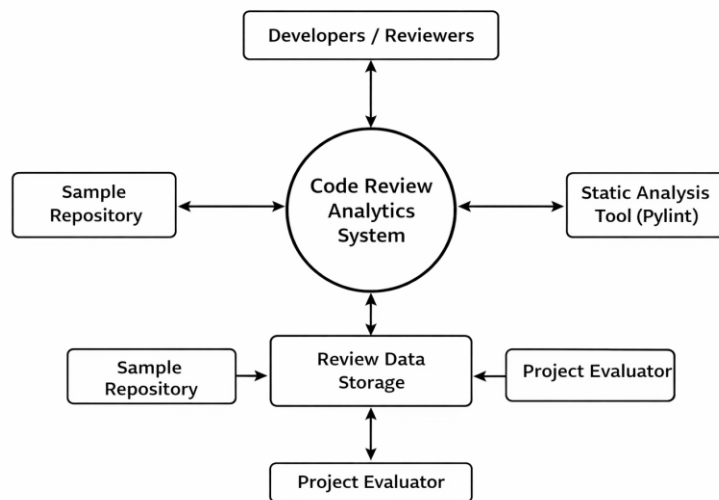
The platform will interface with:

- Sample software repositories used for conducting code review experiments
- Version control systems (e.g., Git-based repositories) for accessing pull request data
- Static analysis tools such as Pylint for automated defect detection

- Local datasets containing review comments, defect logs, and review metrics

The system will collect review-related data, process it through analytical and machine learning modules, and generate comparative reports to evaluate the effectiveness of various code review techniques.

The following context diagram represents the system's position within its environment and its interaction with external entities such as reviewers, repositories, and analysis tools.



## 2.2. Product Functions

The Code Review Analytics System Based on Machine Learning will provide the following major functions:

### 1) Code Review Data Management

- Collection and storage of code review comments
- Recording defect type and severity (Critical, Major, Minor)
- Tracking review time and reviewer information
- Managing multiple review techniques (peer, pull-based, tool-assisted)

### 2) Review Technique Management

- Creation and configuration of review sessions
- Selection of review method (manual, tool-assisted, hybrid)
- Management of review status (active, completed, archived)

### 3) Machine Learning Analysis

- Severity prediction using trained ML classification models
- Identification of defect patterns from review data
- Clustering of defect categories
- Predictive estimation of review effectiveness

### 4) Metrics and Scoring

- Calculation of Weighted Code Quality Score (WQS)
- Measurement of review effectiveness and efficiency
- Comparative evaluation of different review techniques

#### 5) Reporting and Visualization

- Dashboard displaying key performance indicators
- Graphical visualization of defect distribution
- Comparative performance charts of review techniques
- Exportable analytical reports

#### 6) System Administration

- User management and access control
- Configuration of system parameters
- Data backup and recovery management
- System logging and monitoring

### 2.3 User Classes, Characteristics, and Needs

The system will serve the following user classes:

#### 1) Developers / Reviewers

- Individuals performing code reviews
- Basic programming knowledge
- Need to submit review comments and analyze defect findings

##### **Primary Needs:**

Efficient defect recording, review tracking, and performance feedback.

#### 2) Project Evaluators / Faculty

- Academic supervisors and examiners
- Interested in reviewing analysis results

##### **Primary Needs:**

Comparative analytics, performance metrics, and report generation.

#### 3) System Administrator

- Technical personnel managing the system
- Responsible for maintenance and monitoring

##### **Primary Needs:**

User management, data backup, system configuration, and monitoring.

### 2.4 Operating Environment

#### 1) Technical Environment

- Web-based or Python-based analytics platform
- Accessible via modern web browsers
- May support local deployment or cloud hosting

## **2) Hardware Environment**

- Standard desktop/laptop systems
- Server capable of supporting ML processing
- Adequate storage for review datasets

## **2) Software Environment**

- Python programming environment
- Machine Learning libraries (e.g., scikit-learn)
- Database system (e.g., MySQL, PostgreSQL)
- Visualization tools (e.g., Matplotlib, Streamlit)

## **3) User Environment**

- Desktop or laptop devices
- Basic internet connectivity
- Users with varying technical proficiency

## **2.5 Design and Implementation Constraints**

### **1) Technical Constraints**

- Must operate using open-source technologies
- Must support integration with Git-based repositories
- Must function in limited-resource academic environments

### **2) Regulatory Constraints**

- Must ensure protection of stored review data
- Must follow academic integrity guidelines

### **3) Business Constraints**

- Implementation limited to academic timeline
- Basic functionality prioritized over advanced optimization

### **4) User Constraints**

- Must be simple and intuitive
- Must accommodate users with different technical backgrounds

## **2.6 User Documentation**

The following documentation will be developed:

### **1) Online Help System**

- Context-based help for system features
- FAQs

## 2) User Manuals

- Reviewer User Guide
- Administrator Guide
- Project Evaluator Guide

## 2) Training Materials

- Quick reference documentation
- Demonstration slides and screenshots

## 2.7 Assumptions and Dependencies

### Assumptions

1. Developers will provide accurate review data.
2. Adequate computing resources will be available for ML processing.
3. The project will be implemented in academic phases.
4. Stakeholders will validate requirements during development.

### Dependencies

1. Availability of sample software repositories for review experiments.
2. Availability of required machine learning libraries and tools.
3. Access to version control systems for pull-based review data.
4. Institutional approval for experimental evaluation.

## 3. System Features and Functional Requirements

This section describes the main features of the Code Review Analytics System Based on Machine Learning. The system collects code review data, analyzes review effectiveness, uses machine learning for predictive insights, and generates metrics to evaluate various code review practices.

Each requirement outlines how the system should operate and what services it must offer to users.

### 3.1. User Management and Authentication

The system will provide secure user management to allow controlled access to the platform.

#### 3.1.1 User Registration

Users (students, developers, or researchers) can create accounts by providing basic details like name, email, and password.

The system will verify user identity through email authentication.

#### 3.1.2 User Roles

The system will support role-based access with these roles:

- Researcher / Developer – Can upload review data, run analysis, and view reports.
- Administrator – Can manage users, system settings, and machine learning configurations.
- The system will restrict access to features based on assigned roles.

### 3.1.3 Authentication and Security

- The system will implement secure login using encrypted passwords.
- Session management will help prevent unauthorized access.
- The system will keep logs of user activity for auditing.

## 3.2 Code Submission and Review Data Management

This module handles the collection and storage of code review information.

### 3.2.1 Code Submission

Users can upload source code files or repository links for review analysis. The system will support commonly used programming languages like Python and Java.

### 3.2.2 Review Data Entry

Users can manually enter review comments, defect descriptions, and severity levels.

The system will store these review attributes:

- Review technique used (Peer, Pull-based, Tool-assisted)
- Number of defects found
- Review time
- Lines of code reviewed
- Reviewer comments
- Defect type and category

### 3.2.3 Review Data Storage

The system will securely store review data in a structured database. Users can retrieve and modify stored review data.

## 3.3 Defect Analysis and Classification

The system will analyze found defects and classify them based on severity levels.

### 3.3.1 Defect Severity Classification

The system will categorize defects into:

- Critical
- Major
- Minor

Administrators can configure severity weight values used in evaluation metrics.

### 3.3.2 Defect Categorization

Defects will be categorized into types such as:

- Security
- Performance
- Design
- Logical
- Style

The system will support grouping similar defects for trend analysis.

## 3.4 Machine Learning Module

The Machine Learning module is responsible for predictive analysis and pattern recognition.

### 3.4.1 Severity Prediction

The system will use a supervised machine learning model to predict defect severity based on historical review data.

It will use input features like:

- Review comment text
- Lines of code changed
- Review duration
- Module name

The system will output predicted severity labels.

### 3.4.2 Defect Clustering

Unsupervised learning techniques will cluster similar defects. Clustering will help identify major problem areas in the software system.

### 3.4.3 Review Effectiveness Prediction

The system will use regression analysis to estimate the expected number of defects found based on review effort and technique.

### 3.4.4 Model Training and Evaluation

The system will allow retraining of machine learning models with updated datasets.

It will compute performance metrics like:

- Accuracy
- Precision
- Recall

## 3.5 Review Effectiveness Evaluation

This module offers a quantitative assessment of code review practices.

### 3.5.1 Weighted Code Quality Score (WQS) Calculation

The system will calculate the Weighted Code Quality Score (WQS) using severity-based weight values. The WQS will help compare different review techniques objectively.

### 3.5.2 Comparative Analysis

The system will compare review techniques based on:

- Total defects found
- Severity distribution
- Review effort
- WQS value

The system will generate comparative reports.

## 3.6 Analytics and Visualization

The system will provide an interactive dashboard to display analysis results.

### 3.6.1 Graphical Visualization

The system will create charts and graphs that represent:

- Defect distribution by severity
- Technique comparison
- Review time versus defects found
- Clustering results

### 3.6.2 Trend Analysis

The system will identify trends and recurring defect patterns across review cycles.

## 3.7 Reporting Module

The system will allow users to generate detailed analytical reports. It will support exporting reports in formats like PDF and CSV.

Reports will include:

- Summary statistics
- WQS results
- Machine learning predictions
- Comparative analysis

## 3.8 System Administration

The system will provide administrative tools to:

- Manage user accounts

- Configure severity weights
- Adjust machine learning parameters
- Monitor system performance

The system will maintain logs for auditing and monitoring.

## 4. External Interface Requirements

### 4.1. User Interfaces

- The system shall provide user interfaces for the following user types:
- **1) Developers / Reviewers**
  - Interface for submitting code review data and comments
  - Dashboard for viewing detected defects and severity classification
  - Visualization panel for comparing review techniques (peer, pull-based, tool-assisted)
  - Notifications for review results and analysis updates
- **2) Project Evaluators / Faculty**
  - Interface for viewing comparative analysis reports
  - Access to graphical dashboards showing WQS and effectiveness metrics
  - Export functionality for reports and evaluation results
- **3) Administrators**
  - User management interface (enable/disable users)
  - System configuration and dataset management interface
  - Monitoring interface for review data processing and analytics modules
- **4) Guest Users**
  - Read-only interface to view sample analysis results and system overview
  - Prompt to log in for accessing detailed analytics and data entry features

All user interfaces shall adhere to the following requirements:

- User-friendly and interactive dashboard-based UI
- Responsive design for desktop and laptop devices
- Consistent navigation and layout across modules
- Graphical visualization of review metrics (charts, comparisons, trends)
- Context-sensitive help and guidance for new users.

## 4.2. Hardware Interfaces

The system shall interface with the following hardware components:

### 1) Server Infrastructure

- Application server for analytics processing
- Database server for storing review data and metrics
- Backup storage for datasets and generated reports

### 2) Client Devices

- Desktop computers
- Laptops

The system shall minimize hardware dependency and support standard computing devices available in academic environments.

## 4.3. Software Interfaces

The system shall interface with the following software components:

### 1) Version Control Systems

- Integration with Git-based repositories for accessing code review data

### 2) Static Analysis Tools

- Integration with tools such as Pylint or similar analyzers for automated defect detection

### 3) Machine Learning Libraries

- Interface with ML libraries (e.g., scikit-learn) for severity prediction and pattern analysis

### 4) Data Visualization Tools

- Integration with visualization libraries for generating graphs and dashboards

All software interfaces shall define clear data formats, protocols, and access mechanisms.

## 4.4. Communications Interfaces

The system shall support the following communication interfaces:

### 1) Network Protocols

- HTTP/HTTPS for web-based access to dashboards and reports

### 2) API Communications

- RESTful APIs for data exchange between analytics modules and storage systems
- JSON format for structured data communication

### 3) Data Exchange Formats

- JSON for internal data processing
- CSV for importing/exporting review datasets

All communications shall be secured using authentication and appropriate access control mechanisms.

## 5. Non-Functional Requirements

This section specifies the non-functional requirements for the Code Review Analytics System Based on Machine Learning. These requirements define the quality attributes and operational constraints that the system must satisfy.

### 5.1. Performance Requirements

#### 5.1.1. Response Time

| ID     | Requirement                                                                                        |
|--------|----------------------------------------------------------------------------------------------------|
| NFR-01 | The system SHALL load dashboards and analysis pages within 3 seconds under normal load conditions. |
| NFR-02 | The system SHALL process review data inputs within 2 seconds for standard operations.              |
| NFR-03 | The system SHALL complete machine learning severity prediction within 5 seconds per review batch.  |
| NFR-04 | The system SHALL generate analytical reports within 10 seconds for standard datasets.              |
| NFR-05 | The system SHALL not experience more than 30% performance degradation under peak academic usage.   |

#### 5.1.2. Throughput

| ID     | Requirement                                                                              |
|--------|------------------------------------------------------------------------------------------|
| NFR-06 | The system SHALL support at least 50 concurrent users during normal academic operations. |
| NFR-07 | The system SHALL process at least 200 review entries per session.                        |

|        |                                                                                          |
|--------|------------------------------------------------------------------------------------------|
| NFR-08 | The system SHALL support analysis of at least 10,000 defect records without failure.     |
| ID     | Requirement                                                                              |
| NFR-06 | The system SHALL support at least 50 concurrent users during normal academic operations. |

### 5.1.3. Resource Utilization

|        |                                                                                      |
|--------|--------------------------------------------------------------------------------------|
| ID     | Requirement                                                                          |
| NFR-09 | The system SHALL utilize no more than 80% CPU during peak analysis tasks.            |
| NFR-10 | The system SHALL utilize no more than 80% available memory during normal operations. |
| NFR-11 | The system SHALL optimize database queries to reduce response delays.                |
| NFR-12 | The system SHALL implement efficient data processing to minimize storage usage.      |
| ID     | Requirement                                                                          |

## 5.2. Security Requirements

### 5.2.1. Authentication and Authorization

|        |                                                                                      |
|--------|--------------------------------------------------------------------------------------|
|        | Requirement                                                                          |
| NFR-09 | The system SHALL utilize no more than 80% CPU during peak analysis tasks.            |
| NFR-10 | The system SHALL utilize no more than 80% available memory during normal operations. |
| NFR-11 | The system SHALL optimize database queries to reduce response delays.                |
| NFR-12 | The system SHALL implement efficient data processing to minimize storage usage.      |
| ID     | Requirement                                                                          |
| NFR-09 | The system SHALL utilize no more than 80% CPU during peak analysis tasks.            |
| NFR-10 | The system SHALL utilize no more than 80% available memory during normal operations. |

### 5.2.2. Data Protection

|    |             |
|----|-------------|
| ID | Requirement |
|----|-------------|

|        |                                                                                        |
|--------|----------------------------------------------------------------------------------------|
| NFR-17 | The system SHALL encrypt sensitive data in transit using secure protocols (HTTPS/TLS). |
| NFR-18 | The system SHALL protect stored data against unauthorized access.                      |
| NFR-19 | The system SHALL implement secure backup mechanisms for review data.                   |

### 5.3. Reliability and Availability

#### 5.3.1. Availability

| ID     | Requirement                                                                      |
|--------|----------------------------------------------------------------------------------|
| NFR-20 | The system SHALL maintain at least 95% availability during academic usage hours. |
| NFR-21 | The system SHALL schedule maintenance during low-usage periods.                  |

#### 5.3.2. Fault Tolerance

| ID     | Requirement                                                               |
|--------|---------------------------------------------------------------------------|
| NFR-22 | The system SHALL recover automatically from minor system failures.        |
| NFR-23 | The system SHALL maintain data integrity in case of unexpected shutdowns. |

### 5.4. Usability and Accessibility

#### 5.4.1. User Interface

| ID     | Requirement                                                                             |
|--------|-----------------------------------------------------------------------------------------|
| NFR-24 | The system SHALL provide an intuitive and consistent user interface.                    |
| NFR-25 | The system SHALL provide graphical visualization of analytics results.                  |
| NFR-26 | The system SHALL provide clear error messages without exposing internal system details. |
| NFR-27 | The system SHALL provide user documentation and guidance.                               |

### 5.5. Maintainability and Portability

#### 5.5.1. Maintainability

| ID | Requirement |
|----|-------------|
|----|-------------|

|        |                                                                            |
|--------|----------------------------------------------------------------------------|
| NFR-28 | The system SHALL follow modular architecture principles.                   |
| NFR-29 | The system SHALL include proper documentation and comments in source code. |
| NFR-30 | The system SHALL support future model updates without major redesign.      |

#### 5.5.2. Portability

| ID     | Requirement                                                               |
|--------|---------------------------------------------------------------------------|
| NFR-31 | The system SHALL operate on Windows and Linux environments.               |
| NFR-32 | The system SHALL support deployment on local or cloud-based environments. |

### 5.6. Legal and Compliance Requirements

#### 5.6.1. Regulatory Compliance

| ID       | Requirement                                                                                  |
|----------|----------------------------------------------------------------------------------------------|
| NFR-118. | The system SHALL comply with all applicable Palestinian labor laws and regulations.          |
| NFR-119. | The system SHALL incorporate GDPR principles as best practice for data protection.           |
| NFR-120. | The system SHALL comply with accessibility regulations and standards.                        |
| NFR-121. | The system SHALL maintain appropriate records for regulatory compliance and auditing.        |
| NFR-122. | The system SHALL implement mechanisms to stay current with changing regulatory requirements. |

#### 5.6.2. Intellectual Property

| ID       | Requirement                                                                             |
|----------|-----------------------------------------------------------------------------------------|
| NFR-123. | The system SHALL respect intellectual property rights in all content and functionality. |
| NFR-124. | The system SHALL properly license all third-party components and libraries.             |
| NFR-125. | The system SHALL provide appropriate attribution for third-party content.               |
| NFR-126. | The system SHALL implement mechanisms to prevent copyright infringement by users.       |

#### 5.6.3. Service Level Agreements

| ID | Requirement |
|----|-------------|
|----|-------------|

|          |                                                                                               |
|----------|-----------------------------------------------------------------------------------------------|
| NFR-127. | The system SHALL define and document service level agreements (SLAs) for system availability. |
| NFR-128. | The system SHALL define and document SLAs for incident response and resolution times.         |
| NFR-129. | The system SHALL define and document SLAs for support services.                               |
| NFR-130. | The system SHALL implement monitoring and reporting mechanisms to track SLA compliance.       |
| NFR-131. | The system SHALL define escalation procedures for SLA violations.                             |

## 5.7. Operational Requirements

### 5.7.1. Monitoring and Logging

| ID       | Requirement                                                                               |
|----------|-------------------------------------------------------------------------------------------|
| NFR-132. | The system SHALL implement comprehensive logging for all system components.               |
| NFR-133. | The system SHALL provide real-time monitoring of system health and performance.           |
| NFR-134. | The system SHALL generate alerts for critical system events and performance thresholds.   |
| NFR-135. | The system SHALL maintain log retention policies in compliance with legal requirements.   |
| NFR-136. | The system SHALL provide dashboards for monitoring system status and performance metrics. |
| NFR-137. | The system SHALL implement log aggregation and analysis tools.                            |

### 5.7.2. Backup and Recovery

| ID       | Requirement                                                                |
|----------|----------------------------------------------------------------------------|
| NFR-138. | The system SHALL perform automated backups according to defined schedules. |
| NFR-139. | The system SHALL verify backup integrity through automated testing.        |
| NFR-140. | The system SHALL provide mechanisms for point-in-time recovery.            |
| NFR-141. | The system SHALL document and test restoration procedures.                 |
| NFR-142. | The system SHALL maintain backup history and audit trails.                 |

### 5.7.3. System Administration

| ID       | Requirement                                                                                 |
|----------|---------------------------------------------------------------------------------------------|
| NFR-143. | The system SHALL provide administrative interfaces for system configuration and management. |
| NFR-144. | The system SHALL support role-based access for administrative functions.                    |

|          |                                                                                   |
|----------|-----------------------------------------------------------------------------------|
| NFR-145. | The system SHALL provide tools for user management and support.                   |
| NFR-146. | The system SHALL implement change management procedures for system modifications. |
| NFR-147. | The system SHALL provide mechanisms for content moderation and management.        |
| NFR-148. | The system SHALL support system health checks and diagnostics.                    |

#### 5.7.4. Documentation

| ID       | Requirement                                                                                |
|----------|--------------------------------------------------------------------------------------------|
| NFR-149. | The system SHALL provide comprehensive user documentation for all user roles.              |
| NFR-150. | The system SHALL provide technical documentation for system administrators and developers. |
| NFR-151. | The system SHALL maintain up-to-date system architecture and design documentation.         |
| NFR-152. | The system SHALL provide API documentation for integration partners.                       |
| NFR-153. | The system SHALL document all configuration parameters and their effects.                  |
| NFR-154. | The system SHALL provide troubleshooting guides and known issue documentation.             |

### 5.8. Cultural and Political Requirements

#### 5.8.1. Cultural Considerations

| ID       | Requirement                                                                                                  |
|----------|--------------------------------------------------------------------------------------------------------------|
| NFR-155. | The system SHALL respect cultural norms and sensitivities in the Palestinian context.                        |
| NFR-156. | The system SHALL use appropriate terminology and language for the local context.                             |
| NFR-157. | The system SHALL support local date and time formats, including Hijri calendar references where appropriate. |
| NFR-158. | The system SHALL consider gender sensitivities in user interfaces and communications.                        |

#### 5.8.2. Political Considerations

| ID       | Requirement                                                                                           |
|----------|-------------------------------------------------------------------------------------------------------|
| NFR-159. | The system SHALL use politically neutral terminology in system interfaces and documentation.          |
| NFR-160. | The system SHALL respect the political sensitivities of the region in geographic references and maps. |
| NFR-161. | The system SHALL implement appropriate content moderation policies for politically sensitive content. |

|          |                                                                                                                                                     |
|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
| NFR-162. | The system SHALL ensure equitable access for users across all Palestinian territories, including Gaza and the West Bank (including East Jerusalem). |
|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------|

## 6. Other Requirements

### 6.1. Data Migration

- The system shall support migration of existing experimental datasets and review logs into the new analytics platform. This includes:

#### 1) Data Mapping and Transformation

- Mapping of existing review datasets to the system's internal schema
- Transformation of defect logs, review comments, and metrics into standardized formats
- Validation of migrated data for correctness and completeness

#### 2) Migration Process

- Phased migration of sample repositories and review datasets to avoid disruption
- Testing and verification of migrated data before analytics processing
- Rollback mechanisms in case of migration or compatibility issues

#### 3) Data Cleansing

- Identification and correction of inconsistent or missing review data
- Removal of duplicate defect entries and repeated review records
- Standardization of severity labels (Critical, Major, Minor) and review metrics

6.2. Internationalization Requirements The system shall support internationalization to enable broader usability and future scalability.

#### 1) Localization Framework

- Support for adding additional languages in future versions
- Separation of UI text from program logic for easier translation
- Localization support for charts, labels, and reports

#### 2) Cultural Adaptations

- Support for different date, time, and number formats
- Adaptation of content presentation for diverse academic environments
- Consideration of readability and usability differences across regions

### 3) Regional Settings

- Support for region-specific formatting standards
- Adaptability to different academic evaluation practices
- Compliance with institutional and regional software usage policies

## 6.3. Training Requirements

- The system shall provide training support for different user groups.

### 1) Administrator Training

- System configuration and analytics module management
- User access control and dataset management
- Security monitoring and troubleshooting procedures

### 2) End-User Training

- Training materials for developers and reviewers on using analytics dashboards
- Guidance on entering review data and interpreting WQS results
- Self-learning documentation and tutorials

### 3) Training Delivery

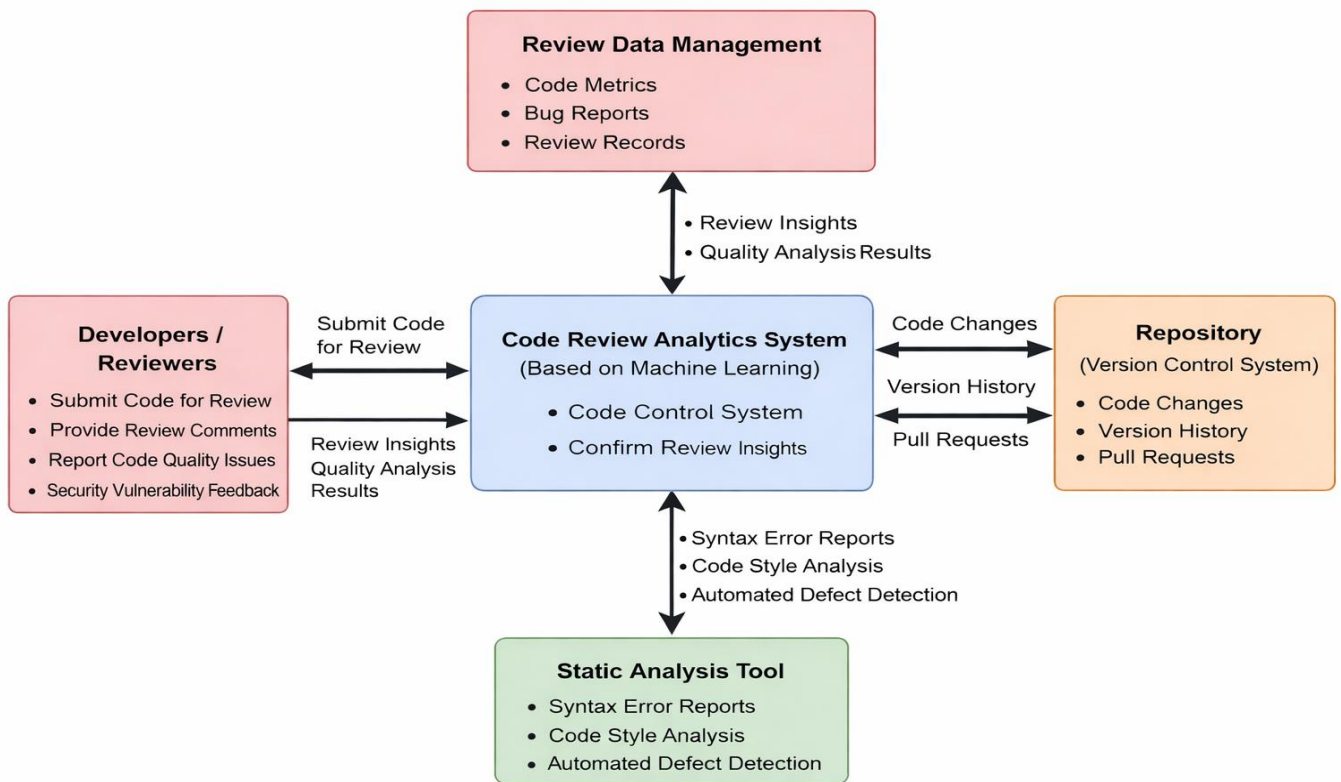
- Online documentation and user manuals
- Demonstration sessions for project evaluators and students
- Continuous training support for future system updates

## 6.4. Appendix A: Analysis Models

This appendix will include the following analysis models:

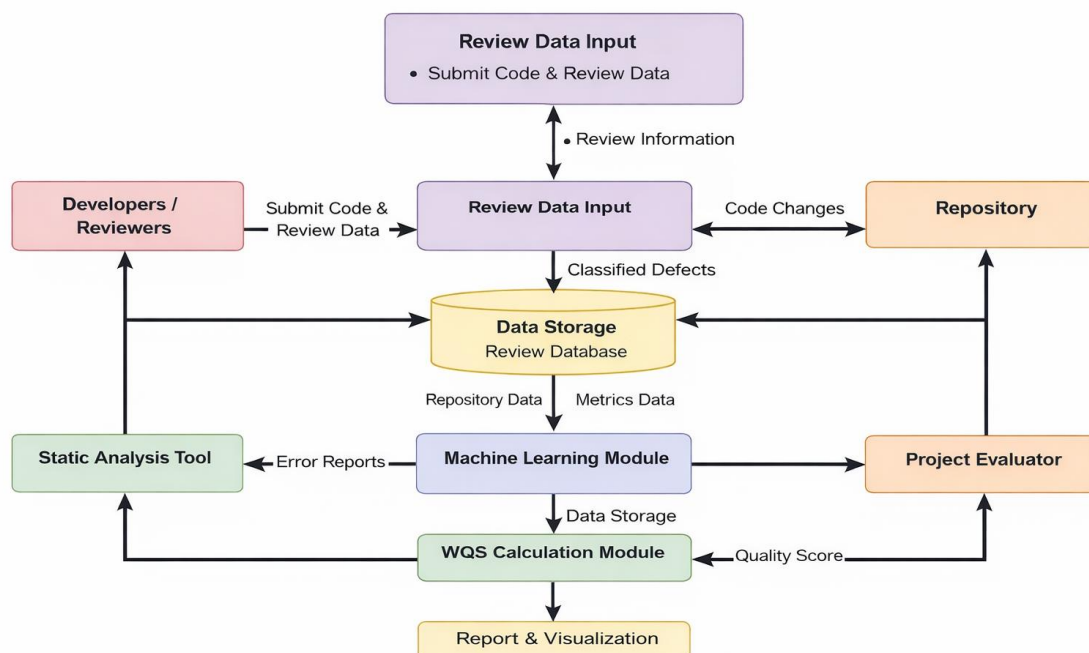
### 1) Context Diagram

- System boundaries and external entities (Level 0)

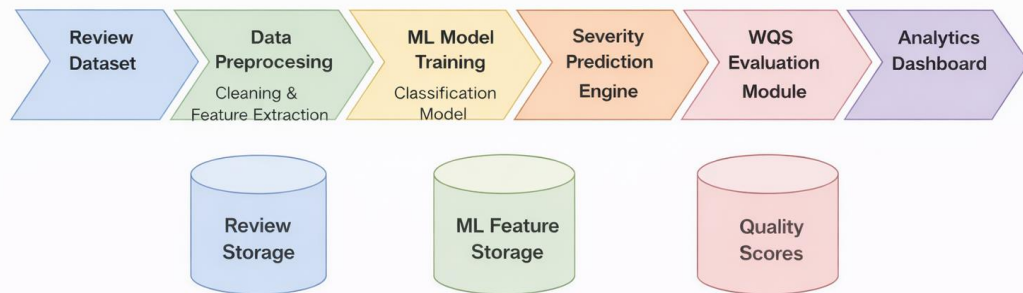


**Code Review Analytics System (Based on Machine Learning)**

- Data flows between the system and external entities (Level 1)

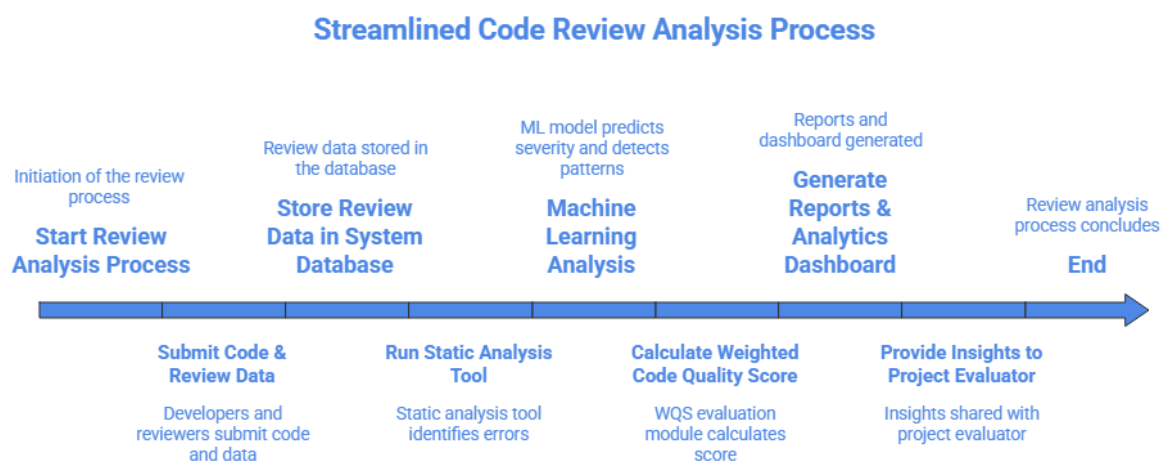


## 2) Machine Learning Engine Component Diagram Diagram



## 3) Process Flow Diagrams

- Code Review Analysis process



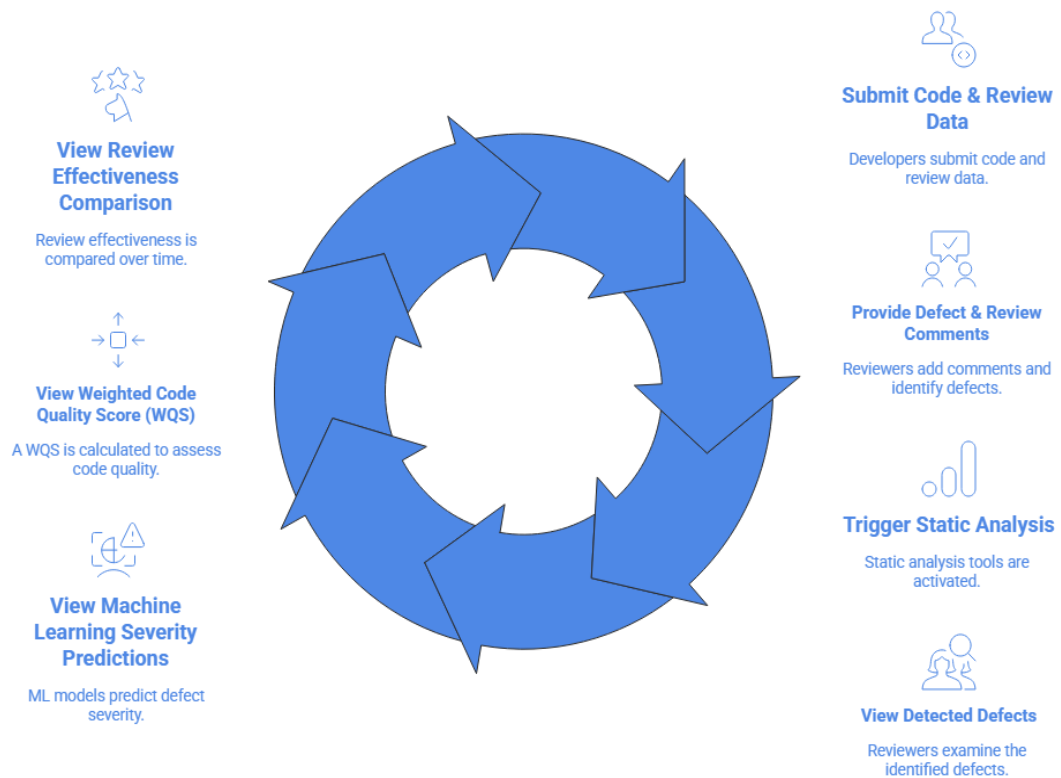
#### 4) Entity Relationship Diagram (ERD) :



#### 5) Use Case Diagrams

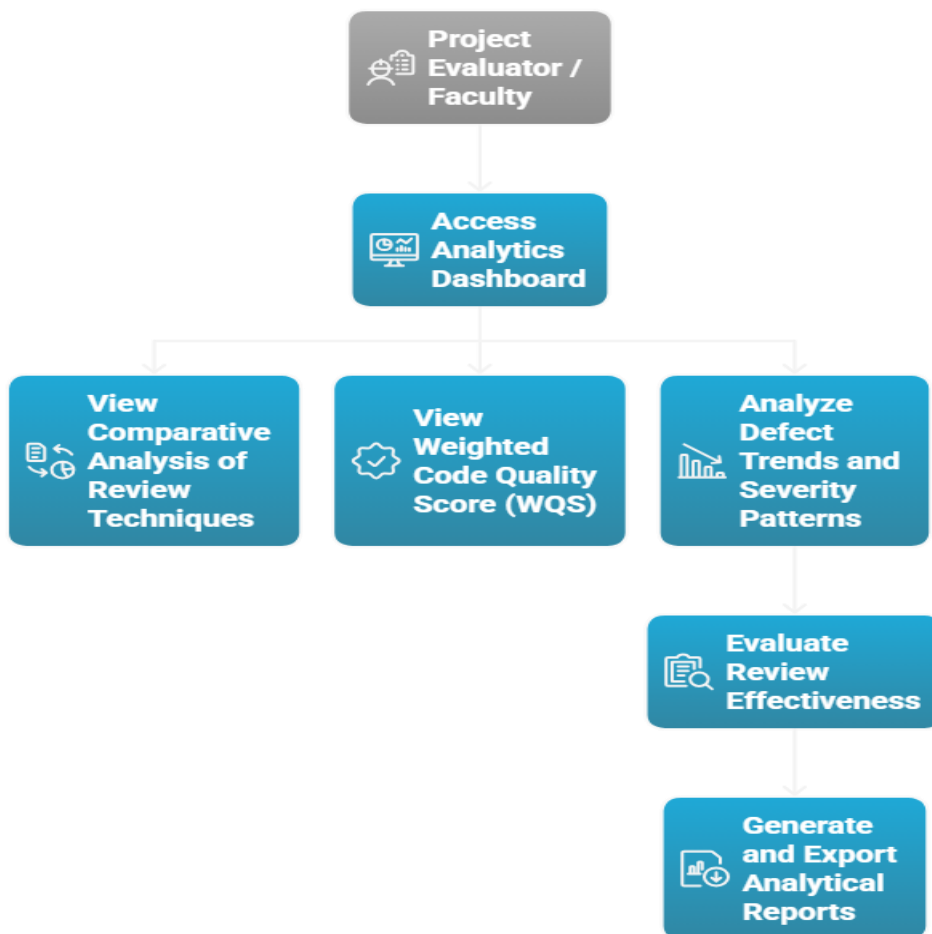
- Developers / Reviewers use case diagram

## Code Review Cycle



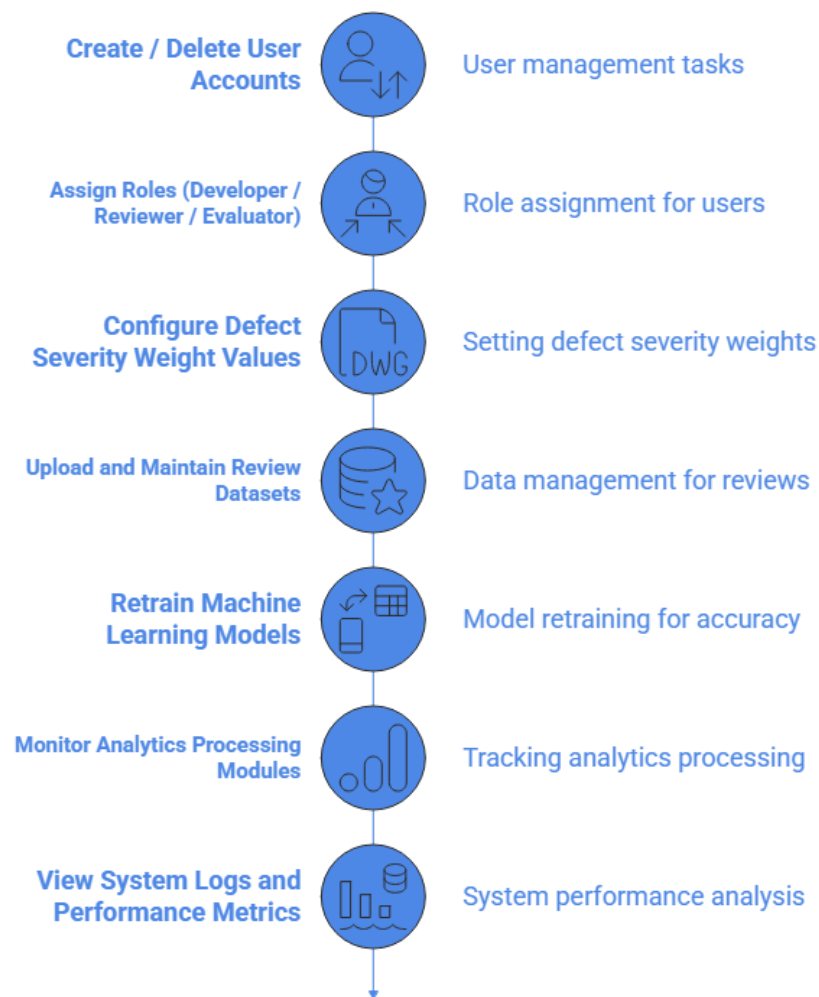
- Project Evaluator Use Case

## Project Evaluation Process



- System Administrator Use Case Diagram

### System Administrator's Key Responsibilities



## 6.5. Appendix B: Issues List

This appendix tracks open issues, decisions, and questions that need to be finalized during the development and implementation of the Code Review Analytics System Based on Machine Learning. These items will be reviewed and updated throughout the project lifecycle.

### 1) Integration with External Tools

- Final specifications for integrating version control repositories (e.g., Git-based repositories)
- Selection and configuration details for static analysis tools (e.g., Pylint or similar analyzers)
- Definition of data exchange formats between the system and external tools

### 2) Machine Learning Model Selection

- Final selection of supervised and unsupervised machine learning algorithms for defect severity prediction and clustering
- Determination of appropriate feature set (review comments, LOC, review time, etc.)
- Selection of model evaluation metrics (Accuracy, Precision, Recall, F1-score)

### 3) Dataset Preparation and Quality

- Identification and collection of suitable review datasets for training and testing
- Data preprocessing requirements such as cleaning, normalization, and feature extraction
- Handling of missing or inconsistent review data

### 4) System Deployment and Hosting

- Decision on deployment environment (local academic lab system or cloud-based hosting)
- Infrastructure requirements for supporting machine learning computation and data storage
- Performance considerations for handling large review datasets

### 5) Evaluation Metrics and WQS Configuration

- Final determination of severity weight values used in Weighted Code Quality Score (WQS) calculation
- Validation of effectiveness metrics used to compare different review techniques
- Definition of threshold values for classifying review effectiveness

### 6) Security and Access Control Considerations

- Definition of user authentication and role-based access control mechanisms
- Protection of stored review data and analytical results
- Logging and monitoring requirements for audit purposes

**Note**

This issues list will be continuously updated as design decisions are finalized and new requirements or challenges are identified during the development and testing phases of the project.

